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RESEARCH ARTICLE



Psychometric properties of the Dysexecutive Questionnaire (DEX) in individuals with a previous cardiac event

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ABSTRACT

This psychometric study aimed to evaluate the psychometric properties of the Dysexecutive Questionnaire (DEX) following a cardiac event, and to examine the relationship between self- and informant-ratings. Approximately 7 months post-cardiac event, 196 participants and their informants ($N = 179$) completed the DEX. Reliability was assessed by confirmatory factor analysis and internal consistency estimates. Construct validity was evaluated using two performance-based neuropsychological tests, the Colour-Word Interference Test and the Trail Making Test (TMT), along with the Hospital Anxiety and Depression Scale (HADS). A one-factor model showed acceptable fit for the self-assessed DEX (RMSEA = 0.062) with excellent internal consistency ($\omega > 0.90$). Slight positive associations were found between the DEX and the performance-based executive function measures, while strong positive associations emerged between the self-assessed DEX and the HADS. Overall, self-assessed and informant-reported scores did not differ significantly (self-assessed DEX

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median = 11, $Q_1-Q_3 = 5.75-20$; informant-reported DEX median = 11, $Q_1-Q_3 = 4-19$, $p = 0.924$). However, participants with severe executive dysfunction (TMT-B z-scores ≤ -2) rated their own symptoms as less severe than informants ($p = .001$), indicating impaired self-awareness in a subgroup of participants. In conclusion, the DEX provides a useful perspective into general executive dysfunction in daily life but should be complemented with performance-based measures.

Trial registration: [ClinicalTrials.gov identifier: NCT03543371](https://clinicaltrials.gov/ct2/show/study/NCT03543371).

Introduction

Individuals with previous cardiac events, such as myocardial infarction (MI) and out-of-hospital cardiac arrest (OHCA), show variability in cognitive functioning, particularly in executive function (EF; Blennow Nordström et al., 2023; Byron-Alhassan et al., 2021; Mion et al., 2023; Zook et al., 2022). Executive functions (EFs) refer to top-down mental processes – including inhibitory control, working memory and cognitive flexibility – that are critical for complex, goal-directed behaviours such as problem solving, reasoning, and planning (Diamond, 2013). EF closely relates to processing speed, which may facilitate or hinder the demonstration of EF abilities, especially in older adults (Diamond, 2013; Salthouse, 1992).

Cardiovascular burden is generally associated with an increased risk of cognitive impairment as a consequence of atherosclerosis, arteriosclerosis, and other vascular risk factors (Qiu & Fratiglioni, 2015). Long-term cardiovascular burden often precedes MI and has been linked to poorer performance on tasks measuring attention, processing speed, EF, learning and memory (Stefanidis et al., 2018). OHCA may result in acquired brain injury due to hypoxic-ischemic injury, leading to further cognitive decline (Blennow Nordström et al., 2023; Howell et al., 2013). Cognitive impairments following OHCA include memory, attention, processing speed, EF and language (Byron-Alhassan et al., 2021; Zook et al., 2022). Lower EF abilities in particular are associated with poorer treatment adherence (Gaalema et al., 2019; Insel et al., 2006), highlighting the need for reliable and convenient assessment tools to capture executive difficulties in affected clinical populations.

One example of a commonly used rating scale for executive problems is the Dysexecutive Questionnaire (DEX; Burgess et al., 1996), originally developed as a tool to sample common complaints among patients with acquired brain injury. The DEX is used quantitatively in both clinical practice and research in a variety of patient groups due to its brevity, ease of administration, and earlier research showing promising ecological validity (Burgess et al., 1998). The 20-item rating scale is aimed to assess symptoms of executive dysfunction related to emotions, behaviours, motivation, and cognition (Burgess et al., 1996), and is available

both as a self-assessment (DEX-Self) and in an informant-rating format (DEX-Informant).

Research on the psychometric properties of DEX is limited. Although existing studies have focused on different populations (e.g., acquired brain injury, psychiatric, non-clinical), none have examined its reliability and validity in individuals with cardiovascular risk. Analyses regarding the latent structure of the DEX have yielded inconsistent results, with different methodologies and sample types producing factor solutions ranging from one to five (Burgess et al., 1998; Hellebrekers et al., 2017; McGuire et al., 2014; Pedrero-Pérez et al., 2015; Simblett & Bateman, 2011). Even studies employing methods appropriate for ordinal categorical measures have resulted in different factor structures: Simblett and Bateman (2011) concluded that a three-factor model best fits the DEX when looking at a sample of individuals with acquired brain injury, whereas Pedrero-Pérez et al. (2015) supported a unidimensional solution in both a non-clinical sample and an outpatient sample receiving treatment for substance use-related problems.

With regards to construct validity, findings on associations between DEX scores and performance-based test results have been mixed, varying between no associations to moderate associations (Bennett et al., 2005; Boelen et al., 2009; Chaytor et al., 2006). Population effects may have influenced these variable findings according to Evans et al. (1997), who found significant associations between performance-based assessments and the DEX in a brain-injured sample, but not in patients with schizophrenia. Furthermore, anxiety and depression have both been associated with greater impairment of EF as measured by the DEX, though these relationships have only been scarcely explored (Azouvi et al., 2015; Gerstorf et al., 2008; Shaw et al., 2015). Depression and anxiety are common among MI and OHCA survivors and are thought to result both from physiological changes and of psychosocial and functional challenges that follow the cardiac event (Chong et al., 2025; Yaow et al., 2022). Symptoms of anxiety, depression and executive dysfunction may overlap, e.g., problems with decision making could be due to excessive and persistent anxiety, or anhedonia and indecisiveness as in major depressive disorder, or impaired initiation and higher-order EFs. Moreover, EF problems might not only result from neurological damage; some studies report that anxiety and depressive disorders could predispose people to future decreased EF (Hammar et al., 2022; Zainal & Newman, 2022). Hence, accurate characterization is essential for delivering individualized rehabilitation and may enhance understanding of the latent structure of the DEX.

A handful of studies have examined the discrepancy between self- and informant-rated DEX scores as a means of assessing self-awareness. Some research suggests that individuals with acquired brain injury tend to underreport the severity of their EF impairment as measured by the DEX (Barker et al., 2011; Burgess et al., 1998; McGuire et al., 2014), while other studies have not found

any discrepancies in self- and informant-assessed total scores (Boelen et al., 2009; Chan & Bode, 2008; Lesimple et al., 2020). Prior research shows that OHCA survivors may underreport symptom severity as measured by other assessments (van Gils et al., 2022), while not much is known about potential issues with self-awareness of cognitive changes in survivors of acute MI. Capturing impaired self-awareness of EF is however of value as greater self-awareness is related to better rehabilitation outcomes and increased functional independence (Ownsworth & Clare, 2006; Robertson & Schmitter-Edgecombe, 2015; Vilalobos et al., 2020).

Because the ecological validity of EF measurements may vary depending on type and severity of impairment, it is essential to conduct research across populations (Chaytor & Schmitter-Edgecombe, 2003). In the current study, cohorts of individuals with a history of MI or OHCA were merged to form a sample with cardiovascular burden that is sufficiently sized for validation purposes (Clark & Watson, 1995).

In our combined sample of individuals with a previous cardiac event (i.e., MI or OHCA), this study aims to:

- (1) Evaluate the hypothesized one-factor structure (structural validity) and the internal consistency reliability of the DEX.
- (2) Assess construct validity by investigating the associations between the DEX and neuropsychological EF test results, as well as between the DEX and symptoms of anxiety and depression.
- (3) Examine the relationship between self- and informant-ratings on the DEX.

Materials and methods

Participants

Participants with a previous OHCA of cardiac or unknown cause were recruited within the international multicenter trial Targeted Hypothermia versus Targeted Normothermia after Out-of-Hospital Cardiac Arrest (TTM2; Dankiewicz et al., 2021; Lilja et al., 2023). OHCA and acute MI participants in Sweden, Denmark, and the United Kingdom were included within the scope of the sub-study Neuropsychological Outcome After Cardiac Arrest (Blennow Nordström et al., 2020, 2023). Sub-study exclusion criteria were need of interpreter, age 80 years or older, clinical diagnosis with a major neurocognitive disorder (dementia) prior to the cardiac event, ongoing substance abuse, severe frailty, or an inability to meet for a face-to-face examination. An additional inclusion criterion for the present analysis was a completed DEX. Initial results from the sub-study showed that 29% of OHCA and 16% of MI participants exhibited at least borderline to mild impairment defined as z-score performance of ≤ -1 in two or more cognitive domains, but the cohorts only exhibited a slight difference in EF abilities based on performance-based tests (Blennow Nordström et al., 2023).

Procedure

Ethical approval was obtained from the corresponding ethics commissions in all participating countries. Participants received written information about the study as well as of their rights prior to signing a consent form. Approximately 7 months after the cardiac event, participants underwent a full neuropsychological assessment, which included completing the DEX-Self as part of the neuropsychological sub-study. The neuropsychological examination took place in a hospital setting or in the participants' home and was administered by neuropsychologists, psychologists, assistant clinical psychologists, or psychology students under supervision. A battery of neuropsychological tests was administered of which a selection relevant to the current study will be reported. Rating scales were completed briefly before or after the neuropsychological examination. Participants were instructed to allot the DEX-Informant to somebody who knew them well (e.g., a relative, a significant other or a close friend) and to complete the questionnaires privately and confidentially.

Measures

Dysexecutive Questionnaire (DEX). The DEX (Burgess et al., 1996) consists of 20 items that assess the frequency of executive difficulties in daily life. It can be completed by participants rating their own experiences, or by informants providing their perceptions of the participant's difficulties. Items are rated on a five-point Likert-type scale ranging from 0 to 4, corresponding to "Never," "Occasionally," "Sometimes," "Often" and "Very Often." Several studies have since the publication of the DEX found high internal consistency (Cronbach's $\alpha > 0.90$) when the scale was regarded unidimensionally (Bennett et al., 2005; McGuire et al., 2014). No cut-off scores have been established.

Colour-Word Interference Test 3 (CWIT3). The third condition of the CWIT from the Delis-Kaplan Executive Function System (Delis et al., 2001) is primarily a measure of inhibitory control (MacLeod, 1991). The task relies on the Stroop paradigm and requires participants to say the name of the colour with which different coloured words are written, thus foregoing the automatic response of reading the words (Stroop, 1935). Standardized scaled scores (normative $M = 10$, $SD = 3$) adjusted for age were used for completion time (CWIT3-T) and number of errors (CWIT3-E). A cut-off of ≤ 3 scaled scores (≤ -2 z-scores below normative means) was applied as indicative of impairment.

Trail Making Test (TMT). The TMT assesses processing speed, attention, visual scanning and EF (Reitan, 1958; Reitan & Wolfson, 1985). The first part (TMT-A) requires connecting numbered circles in a sequence and primarily measures visuomotor processing speed and attention. The second part (TMT-B) asks of participants to connect numbers in ascending and letters in alphabetical order while alternating between the two tasks, placing additional demands

on cognitive flexibility and working memory. Standardized z-scores (normative $M = 0$, $SD = 1$) of total completion time adjusted for age and educational level were used (Tombaugh, 2004). A cut-off of ≤ -2 z-scores below normative means was applied as indicative of impairment.

Hospital Anxiety and Depression Scale (HADS). The HADS (Zigmond & Snaith, 1983) consists of 14 items, equally divided into two sub-scales assessing symptoms of anxiety (HADS-A) and depression (HADS-D). The items are rated on a four-point Likert-type scale ranging from 0 to 3, resulting in possible score ranges between 0–21. Scores above 7 on each subscale are considered clinically relevant (Bjelland et al., 2002).

Statistical analyzes

Before merging the cohorts, we conducted a Wilcoxon rank-sum test to investigate whether MI and OHCA participants differed in DEX total scores. Variables were examined for the assumptions of parametric analysis visually using Q – Q plots, histograms and by Shapiro–Wilk tests. We calculated demographic and descriptive statistics, presented as means (M) and standard deviations (SD), median and quartile 1 to quartile 3 (Q_1 – Q_3), or numbers (n) and proportion (%). There were no substitutions of missing data.

To evaluate the structural validity of the DEX, we first conducted a confirmatory factor analysis (CFA) based on the one-factor model suggested by Pedrero-Pérez et al. (2015). Although we did tentatively conduct a CFA based on the three-factor model suggested by Simblett and Bateman (2011), a comparison of the two models indicated that the unidimensional solution provided a better fit, as reflected in lower RMSEA values. To handle the ordinal nature of data, the parameters were estimated using the weighted least squares means and variance adjusted (WLSMV) method, which is based on a polychoric correlation matrix. The WLSMV does not assume normally distributed variables and is appropriate for ordinal data such as Likert scales (Li, 2016). Due to the low number of observations of the higher ratings of “Sometimes,” “Often” and “Very often” reported on the DEX in our sample, these categories were collapsed for the CFA. Goodness-of-fit indices followed suggested cut-offs (Hu & Bentler, 1999) and include the χ^2 test with the respective p value; Root Mean Square Error of Approximation (RMSEA, values $< .06$ indicate good fit); Tucker-Lewis Index (TLI, values > 0.95 suggest good fit); Comparative Fit Index (CFI, values > 0.95 suggest good fit), and Standardized Root Mean Residual (SRMR, < 0.08 is considered acceptable fit). Ordinal alpha and ordinal omega (ω) was used as a measure of internal consistency due to the non-parametric nature of the DEX (McNeish, 2018). Cronbach’s alpha (α) was additionally calculated to allow for comparisons to previous studies.

Spearman’s rho (r_s) was used to assess associations between the DEX, the performance-based tests TMT and CWIT3, and the HADS. Correlation coefficients (r_s) ranging from 0.10–0.29 were considered small, 0.30–0.49 as medium, and

0.50 or higher as strong (Cohen, 1988). A Bonferroni correction was applied to account for multiple comparisons.

Self- and informant rater agreement was assessed by Wilcoxon signed-rank test for matched samples. A score as a proxy for potential self-awareness was calculated by subtracting total scores on the DEX-Self from those on the DEX-Informant, akin to previous studies (Barker et al., 2011; Chan, 2001). Participants were then split into two groups based on standardized TMT-B scores being equal to or below a z-score of -2 , indicative of EF impairment of clinical significance. The TMT-B was chosen for stratification due to prior studies showing sensitivity in capturing EF impairment in the target population (Blennow Nordström et al., 2023). A Wilcoxon rank sum test was performed to compare the groups based on their self-awareness scores on the DEX.

All analyzes were performed with R Statistical Software (v4.1.1; R Core Team, 2021). Confirmatory factor analyzes (CFA) were conducted using the Lavaan package (v0.6-13; Rosseel, 2012). We applied an alpha of <0.05 , two-sided, for all analyzes.

Results

Sample characteristics

One hundred and six individuals with prior OHCA and 90 individuals with prior MI completed the DEX. In conclusion, 196 participants completed the DEX, together with their informants ($N = 179$). See flowchart for details on participant inclusion (Supplementary Figure). Demographic details are outlined in Table 1.

With regards to days at hospital after the cardiac event, 51% ($n = 100$) spent 10 days or more in care and 43% ($n = 84$) were discharged after less than 10 days with missing data accounting for the remaining 6% ($n = 12$). At the 7 month follow up, 99% ($n = 194$) of participants were living at home. With regards to occupational status, 48% ($n = 94$) were working and 35% ($n = 68$) were retired due to age. According to self-reported health queries, 70% ($n = 137$) of participants were receiving treatment for hypertension and 15% ($n = 30$) had diabetes.

Descriptive statistics

There was no significant difference between MI and OHCA participant groups on neither the DEX-Self ($W = 5208$, $p = 0.270$) nor the DEX-Informant ($W =$

Table 1. Demographic characteristics.

Participants	Total ($N = 196$)
Age in years, M (SD, Min–Max)	62.45 (10.0, 39–78)
Sex, n (%)	
Female	22 (11)
Male	174 (89)
University-level education with or without degree, n (%)	68 (35)
Time since heart event in days, M (SD)	266 (55)

Note: M = mean; SD = standard deviation.

4267, $p = 0.243$). None of the variables displayed normal distributions. Neuropsychological and questionnaire-based test results are presented in Table 2. M and SD are presented to allow for comparisons to previous studies. At the group level, median scores on the TMT and the CWIT did not indicate clinical impairment. However, some participants demonstrated executive impairment, as reflected by the TMT-B (Q_1 z-score = -1.33). HADS scores did not suggest significant distress in the cohort as a whole. For a visual representation of total score distributions for the DEX-Self and the DEX-Informant, see Figure 1. Lowest possible (zero) scores amounted to $<15\%$ for both the DEX-Self ($n = 11$) and the DEX-Informant ($n = 16$). Item-level statistics of the DEX-Self are presented in Table 3. For female participants ($n = 22$), DEX-Self scores were Mdn (Q_1 – Q_3) = 11 (6–18), and for male participants ($n = 174$), Mdn (Q_1 – Q_3) = 11 (5–20). With respect to age, for participants under 50 years ($n = 25$), DEX-Self scores were Mdn (Q_1 – Q_3) = 15 (5–21), and for those aged 50 and above ($n = 171$), Mdn (Q_1 – Q_3) = 11 (6–20).

Structural validity and reliability of the DEX

The initial CFA of the hypothesized one-factor model showed questionable model fit, as none of the goodness-of-fit indices reached acceptable levels, although they were close: RMSEA = 0.071, CFI = 0.94, TLI = 0.93 and SRMR = 0.09. An inspection of the residual correlation matrix and modification indices revealed a problem with correlated error variances between item 13 and 20. Since both items indicate a lack of concern or indifference towards one's own conduct, the residual variances were allowed to correlate in a revised model. The revised model demonstrated improved model fit according to all goodness of fit indices. Fit indices all approached levels of good fit with RMSEA = 0.062,

Table 2. Descriptive statistics of neuropsychological tests and questionnaires ($N = 196$), including number (n) and proportion (%) of participants who scored above or below the established cut-offs of ≤ -2 z-scores for the TMT and the CWIT, and >7 on subscales of the HADS.

	Median (Q_1 – Q_3)	M (SD)	Indicated clinical impairment, n (%)
DEX-Self	11 (5.75–20)	13.27 (10.13)	NA
DEX-Informant	11 (4–19)	13.64 (11.83)	NA
TMT-A	0.36 (-0.52 – 0.89)	0.02 (1.56)	9 (5%)
TMT-B	-0.11 (-1.33 – 0.88)	-0.73 (3.09)	33 (17%)
CWIT3-T	11 (8–12)	9.74 (3.46)	21 (11%)
CWIT3-E	12 (10–12)	10.86 (2.51)	7 (4%)
HADS-D	2 (1–5)	3.12 (3.12)	21 (11%)
HADS-A	3 (1–6)	3.74 (3.25)	27 (14%)

Note. Q_1 – Q_3 = quartile 1 to quartile 3; DEX-Self = Self-assessed Dysexecutive Questionnaire, total score; DEX-Informant = Informant-rated Dysexecutive Questionnaire, total score; TMT-A = Trail Making Test A, z-score of total completion time; TMT-B = Trail Making Test B, z-score of total completion time; CWIT3-T = Colour-Word Interference Test condition 3, scaled score of total completion time; CWIT3-E = Colour-Word Interference Test condition 3, scaled score of total errors; HADS-A = Hospital Anxiety and Depression Scale – Anxiety, total score; HADS-D = Hospital Anxiety and Depression Scale – Depression, total score; NA = not applicable.

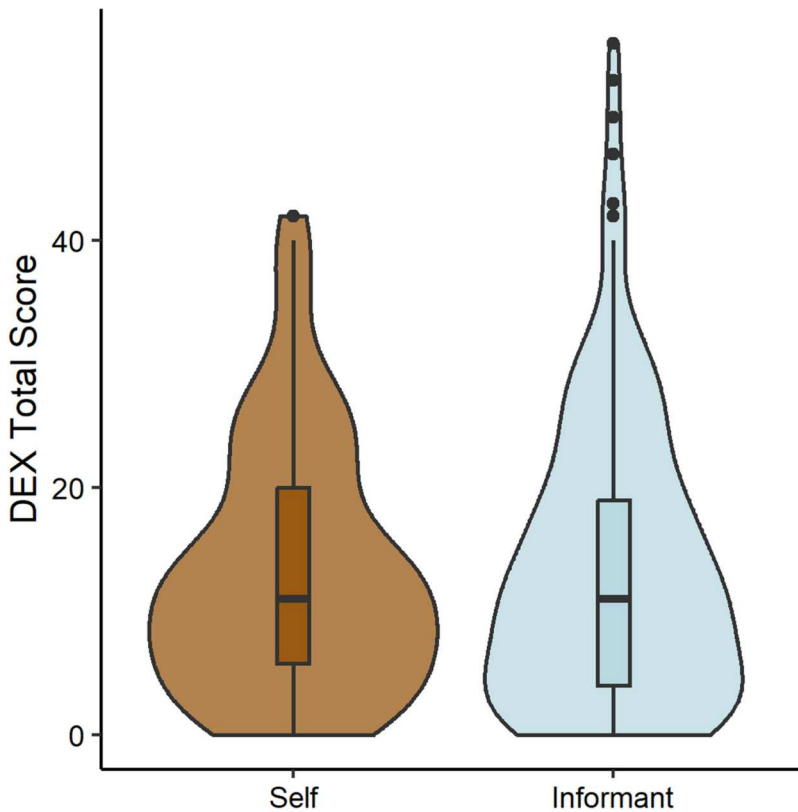


Figure 1. Violin plots showing the distribution of total scores on the DEX-Self ($N = 196$) and DEX-Informant ($N = 179$).

Note. Outliers are marked individually as black circles in the violin plots. DEX-Self, Self-assessed Dysexecutive Questionnaire; DEX-Informant, Informant-rated Dysexecutive Questionnaire.

CFI = 0.95, TLI = 0.95 and SRMR = 0.08. In addition, standardized factor loadings were strong (0.56–0.80) and no Heywood cases were identified. Both ordinal alpha ($\alpha = 0.94$) and ordinal omega ($\omega = 0.94$) indicated excellent internal consistency reliability, like Cronbach's alpha ($\alpha = 0.91$). For further details, see Table 4.

Correlations to neuropsychological tests and measures of anxiety and depression

Correlations are shown in Table 5. On the neuropsychological measures, only the TMT-B correlated significantly with DEX-Self scores ($r_s -0.21$). We found additional weak but significant associations between the DEX-Informant and measures of EF and processing speed ($r_s -0.21$ and -0.25). Associations between the DEX-Self and measures of anxiety and depression were strong ($r_s 0.61$) whereas the associations between these same measures and the DEX-Informant were moderate (HADS-A: $r_s 0.45$; HADS-D: $r_s 0.41$).

Table 3. Item-level descriptives of the Dysexecutive Questionnaire (DEX) including number (*n*) and proportion (%) of participants and informants who answered “Sometimes” or more often to each question.

	DEX-Self		DEX-Informant	
	(N = 196)		(N = 179)	
	Median (Q ₁ –Q ₃)	<i>n</i> (%)	Median (Q ₁ –Q ₃)	<i>n</i> (%)
1. Abstract thinking problems	1 (0–1)	34 (17)	0 (0–1)	33 (19)
2. Impulsivity	1 (0–1)	38 (19)	0 (0–1)	28 (16)
3. Confabulation	0 (0–0)	5 (3)	0 (0–0)	7 (4)
4. Planning problems	0 (0–1)	35 (18)	0 (0–1)	35 (20)
5. Euphoria	0 (0–1)	36 (18)	0 (0–1)	37 (21)
6. Temporal sequencing deficits	0 (0–1)	32 (16)	0 (0–1)	31 (18)
7. Impaired self-awareness and social awareness	0 (0–1)	24 (12)	0 (0–1)	20 (11)
8. Apathy and lack of drive	1 (0–1)	46 (24)	1 (0–2)	49 (28)
9. Disinhibition	0 (0–1)	16 (8)	0 (0–1)	25 (14)
10. Variable motivation	0 (0–1)	24 (12)	0 (0–1)	18 (10)
11. Shallowing of affective responses	1 (0–2)	67 (34)	1 (0–2)	61 (34)
12. Losing temper at slightest	1 (0–1)	39 (20)	1 (0–2)	57 (32)
13. Lack of concern	0 (0–1)	24 (12)	0 (0–1)	21 (12)
14. Perseveration	0 (0–1)	17 (9)	0 (0–1)	24 (14)
15. Restlessness–hyperkinesia	0 (0–1)	32 (16)	0 (0–1)	27 (15)
16. Inability to inhibit responses	0 (0–1)	37 (19)	0 (0–1)	27 (15)
17. Knowing–doing dissociation	0 (0–1)	12 (6)	0 (0–1)	24 (14)
18. Distractibility	1 (0–1)	45 (23)	1 (0–2)	48 (27)
19. Poor decision-making ability	1 (0–1)	38 (19)	1 (0–1)	44 (25)
20. No concern for social rules	0 (0–1)	27 (14)	0 (0–1)	35 (20)

Note: Q₁–Q₃ = quartile 1 to quartile 3.

Relationship between the DEX-Self and DEX-Informant

There was no significant difference between total scores on the DEX-Self and the DEX-Informant based on the Wilcoxon signed-rank test ($p = 0.924$). The discrepancy between DEX-Self and DEX-Informant scores between participants who scored below the threshold for clinically significant EF impairment on the TMT-B ($n = 27$) and those in the normal range ($n = 147$) was significant ($p = .001$), see Figure 2. In those impaired, DEX-Self scores were Mdn (Q₁–Q₃) = 13 (8.5–19), while DEX-Informant scores were Mdn (Q₁–Q₃) = 19 (12.5–28.5). In those unimpaired, DEX-Self scores were Mdn (Q₁–Q₃) = 11 (5–18.5), while DEX-Informant scores were Mdn (Q₁–Q₃) = 10 (4–18.5).

Discussion

This is one of few studies that have evaluated the psychometric properties of the DEX, considering the ordinal nature of data (Azouvi et al., 2015; Chan & Bode, 2008; McGuire et al., 2014; Pedrero-Pérez et al., 2015; Simblett & Bateman, 2011; Simblett et al., 2012). The hypothesized unidimensional latent structure was supported, with good internal consistency reliability. In addition, we found that there were weak but significant associations between DEX and common performance-based measures of EF, while measures of anxiety and depression correlated strongly with the DEX. We found significant discrepancies

Table 4. Confirmatory factor analyzes of the baseline model and revised model of the Dysexecutive Questionnaire (DEX), including standardized factor loadings and error variances of individual items.

	Baseline model	Revised model
1.	0.58 (0.66)	0.58 (0.66)
2.	0.67 (0.55)	0.67 (0.55)
3.	0.67 (0.56)	0.67 (0.55)
4.	0.80 (0.37)	0.80 (0.36)
5.	0.63 (0.61)	0.63 (0.61)
6.	0.69 (0.52)	0.70 (0.52)
7.	0.73 (0.47)	0.73 (0.47)
8.	0.72 (0.48)	0.73 (0.47)
9.	0.69 (0.52)	0.69 (0.52)
10.	0.79 (0.37)	0.80 (0.37)
11.	0.59 (0.65)	0.60 (0.65)
12.	0.64 (0.59)	0.64 (0.59)
13.	0.72 (0.48)	0.67 (0.56) ^a
14.	0.74 (0.46)	0.74 (0.46)
15.	0.57 (0.67)	0.58 (0.67)
16.	0.76 (0.42)	0.76 (0.42)
17.	0.72 (0.49)	0.72 (0.48)
18.	0.75 (0.43)	0.76 (0.43)
19.	0.75 (0.45)	0.75 (0.44)
20.	0.62 (0.61)	0.56 (0.69) ^a
χ^2	335.54	344.96
p -value for χ^2 -test	<0.001	<0.001
RMSEA (90% confidence interval)	0.071 (0.059–0.082)	0.062 (0.050–0.074)
p -value for RMSEA	<0.001	<0.001
CFI	0.94	0.95
TLI	0.93	0.95
SRMR	0.09	0.08
Ordinal alpha	0.94	0.94
Ordinal omega	0.95	0.94
Cronbach's alpha	0.91	0.91

Note: Error variances of items marked with ^a were allowed to correlate, $r = 0.30$.

χ^2 = Chi-square; RMSEA = root mean square error of approximation; CFI = comparative fit index; TLI = Tucker-Lewis index; SRMR = standardized root mean squared residual.

between self- and informant-based ratings in a subgroup of individuals whose performance on executive tests indicated clinical impairment, suggesting possible impaired self-awareness. Total scores on the DEX were lower than expected in our sample of individuals at risk of EF impairment.

Table 5. Spearman's correlations and level of statistical significance between the Dysexecutive Questionnaire (DEX), performance-based test results, and measures of anxiety and depression.

Measure	DEX-self (N = 196)	DEX-informant (N = 179)
TMT-A	−0.15 (0.222)	−0.21 (0.032)
TMT-B	−0.21 (0.020)	−0.25 (0.005)
CWIT3-T	−0.16 (0.184)	−0.23 (0.011)
CWIT3-E	−0.05 (1.000)	−0.14 (0.363)
HADS-A	0.61 (<0.001)	0.45 (<0.001)
HADS-D	0.61 (<0.001)	0.41 (<0.001)

Note: Bonferroni correction was applied to account for multiple comparisons. TMT-A = Trail Making Test A, z-score of total completion time; TMT-B = Trail Making Test B, z-score of total completion time; CWIT3-T = Colour-Word Interference Test 3, scaled score of total completion time; CWIT3-E = Colour-Word Interference Test 3, scaled score of total errors; HADS-A = Hospital Anxiety and Depression Scale – Anxiety; HADS-D = Hospital Anxiety and Depression Scale – Depression.

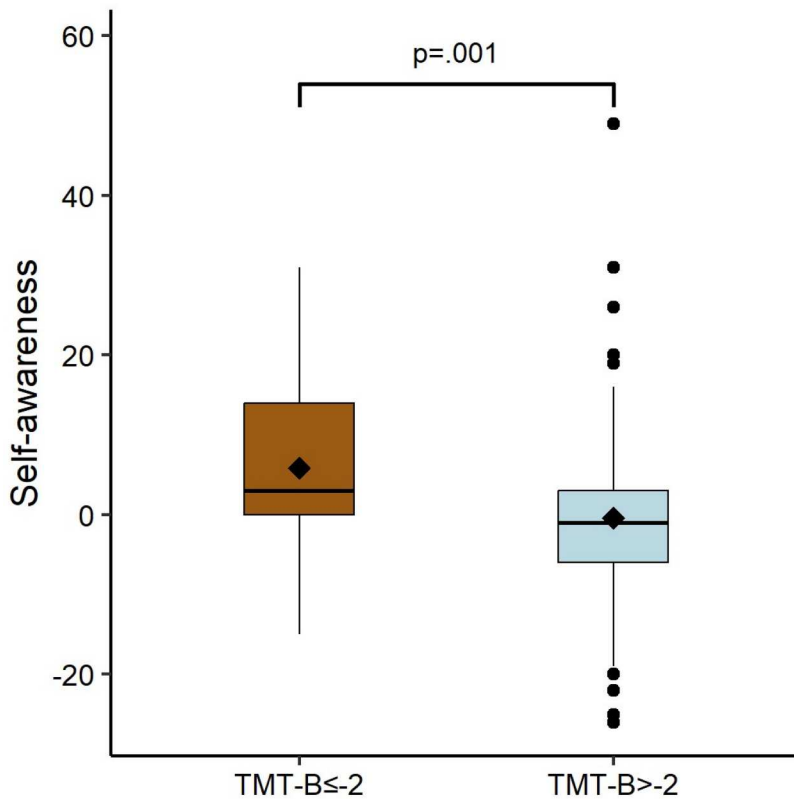


Figure 2. Boxplots demonstrating impaired self-awareness as measured by the DEX based on TMT-B scores, stratified by those scoring equal to or below -2 standardized z-scores ($n = 27$) and those scoring above ($n = 147$) based on total completion time.

Note. Self-awareness was calculated by subtracting scores on the DEX-Self from the DEX-Informant. Diamonds represent means and dots outliers. TMT-B, Trail Making Test B, z-score of total completion time; DEX-Self, Self-assessed Dysexecutive Questionnaire; DEX-Informant, Informant-rated Dysexecutive Questionnaire.

Using a confirmatory factor analytic approach, we found support for the unidimensionality of the DEX. While our baseline model did not exhibit good model fit, goodness-of-fit indices reached established cut-offs when we let the error variances for two items correlate in our revised model. Common problems when finding correlating residuals can be that a given scale is multidimensional, or that items within the scale overlap and are therefore redundant. In our case, we found that the item content for two items (question 13 and 20) was overlapping. When taking this into account and letting the error variances correlate, model fit improved significantly, supporting our notion that the DEX is unidimensional. Furthermore, we did not find any indications that the scale should be considered multidimensionally based on modification indices, factor loadings or correlated errors. Previous literature on the factorial structure of the DEX has however been inconclusive, which could be attributed to sample effects and the use of parametric statistical methods (e.g., Burgess et al., 1998;

Hellebrekers et al., 2017; Simblett & Bateman, 2011). Although our results align with studies which applied statistical analyzes suitable for ordinal and non-normally distributed data (McGuire et al., 2014; Pedrero-Pérez et al., 2015), we do acknowledge that a three-factor solution using Rasch analysis has been found as well (Simblett & Bateman, 2011). Building further evidence on the factorial structure is therefore warranted. Clinicians and researchers may also consider using and evaluating the revised version of the scale which has attempted to improve measurement precision and has shown promising psychometric qualities (DEX-R; Simblett et al., 2017; Wakely et al., 2022). Unfortunately, the DEX-R was not available to us at the time of data collection.

Our finding that ratings on the DEX correlate only weakly with performance on common neuropsychological tests of EF is in keeping with evidence from several clinical groups, including OHCA (Blennow Nordström et al., 2022; Jensen et al., 2024). Individuals with executive difficulties in everyday life – such as those resulting from acquired brain injury – may still perform within the normal range on neuropsychological tests (Gioia & Isquith, 2004; Shallice & Burgess, 1991). Discrepancies between rating-scale and performance-based test results has been attributed to factors such as the lack of consensus of the definition of EF (Jurado & Rosselli, 2007), the absence of process-pure EF measurements (i.e., task-impurity problem; Burgess, 1997), and differing demands of the testing situation and everyday environment (Chaytor & Schmitter-Edgecombe, 2003). Furthermore, rating scales and performance-based assessments may to some extent measure different aspects of EF abilities (Schmitter-Edgecombe et al., 2020). Relatedly, the effectiveness of compensatory strategies and the varying demands of everyday life are thought to shape ratings on self- and informant-reports. The weak but significant correlation between our measure of processing speed and the DEX-Informant may reflect a general mismatch between participants' and informants' view of the challenges that arise in everyday life. In our study, associations between performance-based EF tests and DEX informant-ratings were generally somewhat stronger than those with self-reports, underscoring the importance of including raters with different perspectives when evaluating executive dysfunction.

Unexpectedly, our sample of individuals with a previous cardiac event, along with their informants, reported less severe dysexecutive symptoms on the DEX compared to non-clinical samples from the general population (Burgess et al., 1998; Chan, 2001; Pedrero-Pérez et al., 2015). Given that a large portion of the sample consisted of older individuals – an age group typically associated with EF decline (De Luca et al., 2003; Ferguson et al., 2021) – these low scores were surprising. Although age was not examined in detail due to the limited number of younger participants, we observed that younger individuals had higher DEX scores, indicating slightly more dysexecutive symptoms compared to older participants. One possible explanation is the age-related decline in negative affect, which has been suggested to reduce the perception of

executive dysfunction and lead to lower scores on the DEX (Gerstorff et al., 2008). Younger participants may also be inclined to experience more challenges related to EF impairment due to higher demands in everyday life.

Notably, OHCA survivors tend to report fewer subjective cognitive complaints than would be expected based on performance-based tests (Steinbusch et al., 2017). While our broad group-level analysis did not indicate clinical levels of cognitive impairment, more detailed assessments within this cohort have identified objectively verified EF impairments in a subgroup of individuals, most of whom exhibit mild to borderline deficits (Blennow Nordström et al., 2023). Another proposed explanation is, therefore, a response shift following a life-threatening event, where individuals adjust their perception of cognitive difficulties, leading to fewer reported cognitive complaints (van Gils et al., 2022). This phenomenon may also be relevant for individuals recovering from acute MI and could have had an influence on the total scores of the DEX in our study. An item-level analysis showed the largest floor effects on Item 3 (Confabulation) on both the DEX-Self and the DEX-Informant, aligning with previous findings in other populations (Azouvi et al., 2015; Cederlund et al., 2010). Confabulation may be less commonly encountered than other EF impairment but can be a major and highly disabling symptom. Consequently, the low self- and informant-rated DEX total scores suggest that the scale may have limitations in detecting milder forms of executive dysfunction, as well as certain symptoms such as confabulation, of which individuals are typically unaware.

The DEX-Self showed strong associations with measures of anxiety and depression, while the DEX-Informant showed moderate associations, consistent with previous findings (Azouvi et al., 2015). We hypothesize that the magnitude of the relationships might be explained by item-level content overlap between the HADS and the DEX. For instance, some items on the DEX ask raters to assess feelings of lethargy, unenthusiasm, difficulty in decision-making, and restlessness, symptoms which also characterize anxiety and depression (American Psychiatric Association, 2022). Such analyses were beyond the scope of our study but may warrant further research.

We found no significant difference in total scores between the self- and informant-raters in the overall sample, in contrast with some previous studies. However, those with EF impairment as identified by performance-based tests, rated their difficulties as significantly less severe than informants did. This suggests that the discrepancy between self- and informant scores on the DEX captures impaired self-awareness in those with greater EF impairment. The inconsistencies in previous studies may therefore be due to variabilities in population characteristics. Using both the self- and informant-rated DEX thus shows promise in identifying and evaluating impaired self-awareness over time, which could be valuable in clinical management of patients undergoing neurorehabilitation (Prigatano, 2024).

This study contributes to the existing literature by examining the psychometric properties of the DEX in a novel sample, consisting of individuals with a prior cardiac event. A key strength is the use of statistical methods suited for ordinal and non-normally distributed data. Additionally, the availability of both self- and informant ratings provided insight into potential self-awareness impairments. Combined with performance-based test results, we have shown that EF impairment is related to altered self-awareness of executive dysfunction. To enhance clinical relevance and aid clinicians in selecting appropriate assessments, we included measures of anxiety and depression.

The study also has limitations. First, although most participants allotted the DEX-Informant to family members, we lack detailed knowledge about the informants such as their familiarity with participants' premorbid functioning, which may influence rater accuracy (Barker et al., 2011). Second, additional individual factors such as prior psychiatric or neurodevelopmental diagnoses were not available to us. Third, the sample exhibits an uneven sex distribution, a consequence of men being overrepresented in having cardiovascular disease (Roth et al., 2017). Fourth, because most cognitive recovery occurs within the first three months after cardiac arrest, some individuals may have reported higher levels of impairment if assessed more closely in time following the cardiac event (Byron-Alhassan et al., 2021; Moulaert et al., 2017). Fifth, the sample may not be entirely representative of the population, as participants with cognitive impairment following OHCA may have been slightly underrepresented (Blennow Nordström et al., 2023), and we were not able to investigate measurement invariance and differential item functioning between MI and OHCA participants due to sample size constraints.

The DEX is a brief, easily administered tool for assessing dysexecutive symptoms in adults at risk of EF impairment due to cardiovascular burden. While DEX ratings show some association with objectively measured EF even when symptoms are subclinical, self-reports should be used cautiously, as individuals may both under- and overreport symptom severity. To obtain more comprehensive EF assessments, self-ratings should be used in combination with neuropsychological testing and informant reports. We also recommend screening for anxiety and depression, as both are related to EF as measured by the DEX. The psychometric inconsistencies observed in prior studies highlight the need for caution when using the DEX as a quantitative measure, and the absence of established cut-off scores complicates the interpretation of total scores. Nonetheless, we propose that the DEX is a useful tool that provides a rough estimate of EF impairment and can serve as a starting point for further evaluation.

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No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that will support the findings of this study are available from the TTM2 trial steering group, but restrictions apply to the availability of these data, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of the TTM2 trial steering group.

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